FOUNDATION COURSE

START YOUR

PREPARATION

FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 14

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Recap *of previous lecture*

DETERMINANTS



Topics *to be covered*

"LIMITS"



(1) $\frac{\text{Something}}{0} = \text{N.D}$

(App 1) eg $\frac{40}{5} = 8$

If we have 40 Mangoes & 5 children then each child will get = 8 Mangoes.

eg: $\frac{40}{0} \neq \infty$ → Senseless.

ie when 40 Mangoes are distributed among 0 children then each child get ∞ Mangoes.

(2) Infinite / (Infinity) →

Infinity is not a very large number, it is a presentation of that concept, which is Beyond Imagination.

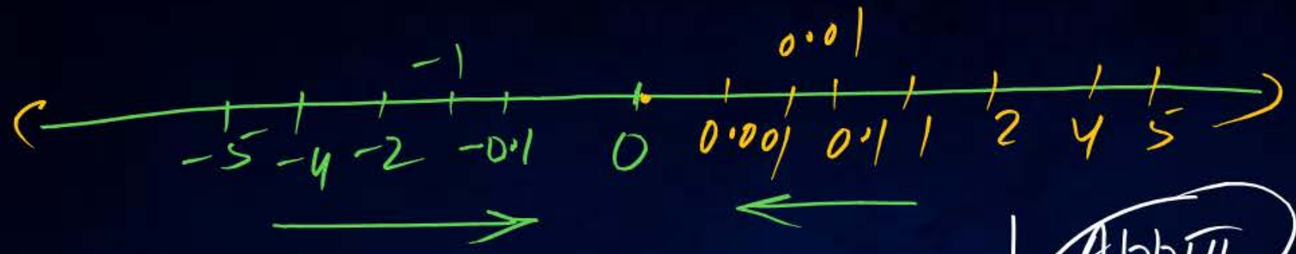
& it is not unique as it depends on the imagination capacity of an Individual.

$\frac{\text{Something}}{0} \cong \infty$ or $-\infty$, $\boxed{\frac{\text{Something}}{\infty} = 0}$

$\boxed{\infty + \infty = \infty}$, $\boxed{\infty \times \infty = \infty}$, $\boxed{\infty^{\infty} = \infty}$ Assumption

while, $\infty - \infty = \text{IND}$ form.

App(2)



$\frac{40}{5} = 8$

$\frac{40}{4} = 10$

$\frac{40}{2} = 20$

$\frac{40}{1} = 40$

$\frac{40}{0.1} = 400$

$\frac{40}{0.01} = 4000$

$\frac{40}{0^+} = \infty$

$\frac{40}{-5} = -8$

$\frac{40}{-4} = -10$

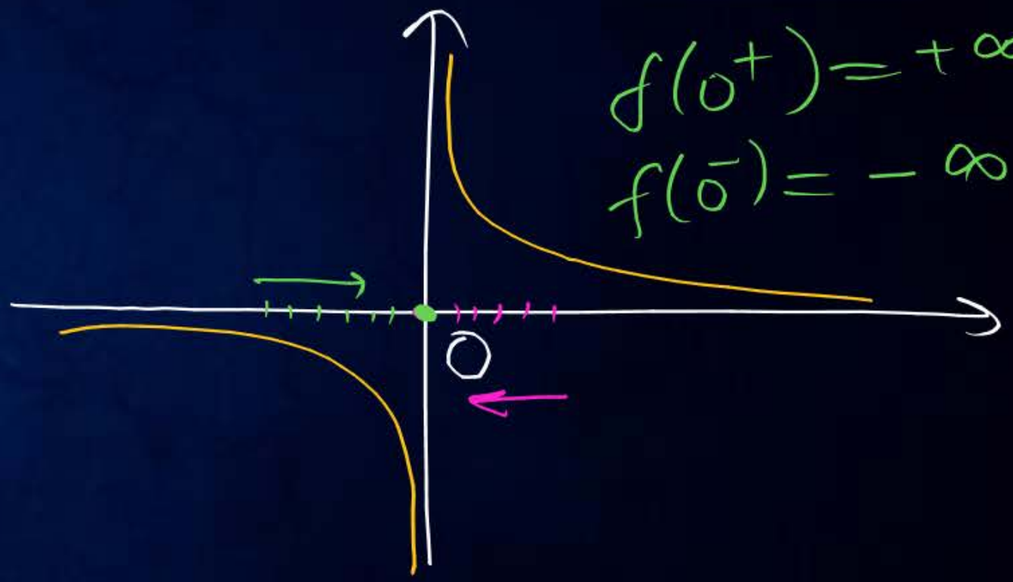
$\frac{40}{-1} = -40$

$\frac{40}{-0.1} = -400$

$\frac{40}{-0.01} = -4000$

$\frac{40}{0^-} = -\infty$

App(III) $y = \frac{1}{x}$ or $f(x) = \frac{1}{x}$



$f(0^+) = +\infty$

$f(0^-) = -\infty$

Indeterminate Form → eg $\frac{0}{0}, \frac{\infty}{\infty}, 0\times\infty, \infty-\infty, 0^0, \infty^0, 1^\infty$

"When given expression has ∞ number of answers then such types of expressions are said to be in (IND) form"

eg $\frac{0}{0} = 1, 2, 3, -1, -5, \frac{1}{4}, \sqrt{7}, \ldots\ldots$ (while, 0 × Number = 0)

Multiple Ans exist

≃ ∞ Ans exist

eg $\frac{\infty}{\infty} = \frac{1/0}{1/0} = \frac{0}{0}$ = IND form

eg: $0\times\infty = 0\times\frac{1}{0} = \frac{0}{0}$ = IND form

eg $(\infty-\infty)$ = After Rationalisation it converts into $\frac{\infty}{\infty}$ form

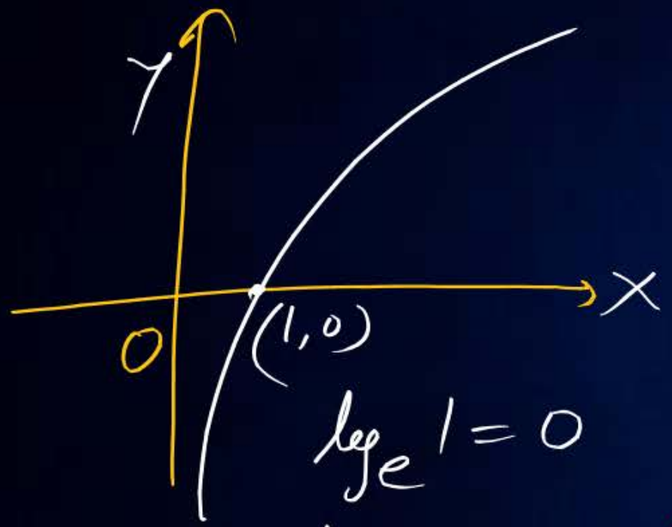
Hence it is also IND form.

$$\lim_{x\to\infty}\left(\sqrt{x^2+x+1}-x\right) \quad (\infty-\infty)$$

$$= \lim_{x\to\infty}\frac{\left(\sqrt{x^2+x+1}-x\right)\left(\sqrt{x^2+x+1}+x\right)}{\sqrt{x^2+x+1}+x} = \lim_{x\to\infty}\left(\frac{x+1}{\sqrt{\quad}+x}\right)$$

$$= \frac{\infty}{\infty} = \text{IND}$$

(*) $y = \log_e x$

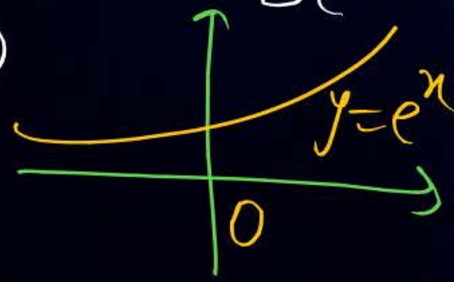


$\log_e 1 = 0$

$\log_e 0 = -\infty$ (=IND)

$\log_e \infty = +\infty$ = (IND)

(*)



$e^{\infty} = +\infty$

$e^{-\infty} = 0$

Let $0^0 = K$

or $K = 0^0$

$\log_e K = \log_e (0^0)$

$= 0 \times \log_e 0$

$= 0 \times (-\infty)$

$= 0 \times \left(\frac{-1}{0}\right)$

$\log_e K = \frac{0}{0}$

$K = e^{\frac{0}{0}}$ = IND form

Let $\infty^0 = K$

or $K = \infty^0$

$\log_e K = \log_e (\infty^0)$

$= 0 \times \log_e \infty$

$= 0 \times \infty$

$= 0 \times \frac{1}{0}$

$\log_e K = \frac{0}{0}$

$K = e^{\frac{0}{0}}$ = IND form

(*) Let $1^{\infty} = k$

or $k = 1^{\infty}$

$\log_e k = \log_e 1^{\infty}$

$= \infty \times \log_e 1$

$= \infty \times 0$

$\log_e k = \frac{1}{0} \times 0 = \frac{0}{0}$

$k = e^{\frac{0}{0}}$ = IND form.

(*) $1^{20} = 1$, $1^{50} = 1$, $1^{27} = 1$, $1^{\infty} \neq 1$

(*) Check the Nature of (1) $\infty^{\infty} = ?$ (2) $0^{\infty} = ?$

(1) Let $\infty^{\infty} = k$

or $k = \infty^{\infty}$

$\log_e k = \log(\infty^{\infty})$

$= \infty \times \log \infty$

$= \infty \times (+\infty)$

$= \infty \times \infty$

$\log_e k = \infty$

$k = e^{\infty} = \infty$

i.e $\infty^{\infty} = \infty$ = N.D.

Let $0^{\infty} = k$

or $k = 0^{\infty}$

$\log_e k = \log(0^{\infty})$

$= \infty \times \log(0)$

$= \infty \times (-\infty)$

$= -(\infty \times \infty)$

$\log_e k = -\infty$

$k = e^{-\infty} = 0$

i.e $0^{\infty} = 0$ = Defined

Limits

Note: $\infty + \infty = N.D$

$\infty^{\infty} = N.D$

$\infty \times \infty = N.D$

$0^{\infty} =$ Defined $(= 0)$

$\underbrace{\frac{0}{0}, \frac{\infty}{\infty}, \infty \times 0, \infty - \infty, 0^0, \infty^0, 1^{\infty}}_{\text{IND form.}}$

Eg $f(x) = \frac{x^2 - 16}{x - 4} = \frac{(x-4)(x+4)}{(x-4)}$

$f(4) = \frac{0 \times 8}{0} = \frac{0}{0} =$ IND form.

So $f(4) =$ DNE (i.e. Exact Value of $f(x)$ at $x = 4$ DNE)

$f(3.999) = \frac{-0.001 \times 7.999}{-0.001} = 7.999 \simeq 8$

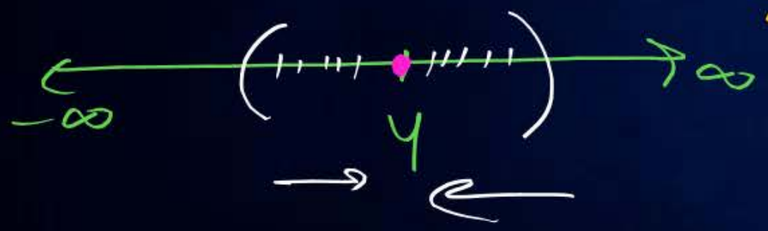
i.e. Value of $f(x)$ in the left Nbd of $4 \cong 8 =$ LHL at $(x = 4)$

$f(4.001) = \frac{+0.001 \times 8.001}{+0.001} = 8.001 \simeq 8$

i.e. Value of $f(x)$ in the Right Nbd of $4 \cong 8 =$ RHL at $(x = 4)$

i.e. $f(x)=\dfrac{x^2-16}{x-4}$

- $f(4)=$ DNE (i.e. Exact Value at $x=4$ DNE)
- LHL $\cong 8$ (Approx Value in the left Nbd of $x=4$)
- RHL $\cong 8$ (" " " Right " ")
- $\lim\limits_{x\to 4} f(x)=8$ (is Approx Value in the Nbd of $x=4$)



When LHL = RHL we say that limit exist.
& when LHL ≠ RHL " " limit DNE

(M-II) $\lim\limits_{x\to 4}\left(\dfrac{x^2-16}{x-4}\right)=\lim\limits_{x\to 4}\left(\dfrac{2x-0}{1-0}\right)=2\times 4=8$

(*) limit exist means it is possible to find unique, const value

Qe If $f(x)=\frac{e^x-1}{x}$ then evaluate it's approx Value in the nbd of $x=0$

{ $\because f(0)=\frac{e^0-1}{0}=\frac{1-1}{0}=\frac{0}{0}$ = IND form i.e exact Value DNE }

(M-I) Approx Value $=\lim_{x\to 0} f(x)=\lim_{x\to 0}\left(\frac{e^x-1}{x}\right)=\lim_{x\to 0}\frac{\left(1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots\right)-1}{x}$

$=\lim_{x\to 0}\left[\frac{x\left(1+\frac{x}{2!}+\frac{x^2}{3!}+\frac{x^3}{4!}+\cdots\right)}{x}\right]=|1.0000|\simeq 1$

(M-II) App Value $=\lim_{x\to 0}\left(\frac{e^x-1}{x}\right)=\lim_{x\to 0}\left(\frac{e^x-0}{1}\right)=e^0=1$

L-Hospital Rule (applicable only for $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form)

$$\lim_{x\to a}\left[\frac{f(x)}{g(x)}\right] \overset{\frac{0}{0}}{=} \lim_{x\to a}\left[\frac{f'(x)}{g'(x)}\right] \overset{\frac{0}{0}}{=} \lim_{x\to a}\left[\frac{f''(x)}{g''(x)}\right] \overset{\frac{0}{0}}{=} \lim_{x\to a}\left[\frac{f'''(x)}{g'''(x)}\right] = \cdots\cdots$$

Advantage:

$$\frac{0}{0} = \underbrace{1, 2, -3, \frac{4}{7}, 15, \frac{2}{3} - - - - -}_{\text{Infinite ans exist.}}$$

i.e. with the help of L-Hosp Rule, we can find unique answer out of ∞ answers.

unique app. Value — ∞ exact Values

Eg: $\lim_{x\to 0}\left(\frac{\sin x}{x}\right) = ?$ Here $f(x)=\frac{\sin x}{x} \Rightarrow f(0)=\frac{0}{0}$ = IND form ie exact value DNE

(M-I) $\lim_{x\to 0}\left(\frac{\sin x}{x}\right) = \lim_{x\to 0}\left[\frac{x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots}{x}\right] = \lim_{x\to 0}\left[1-\frac{x^2}{3!}+\frac{x^4}{5!}-\frac{x^6}{7!}+\cdots\right]$

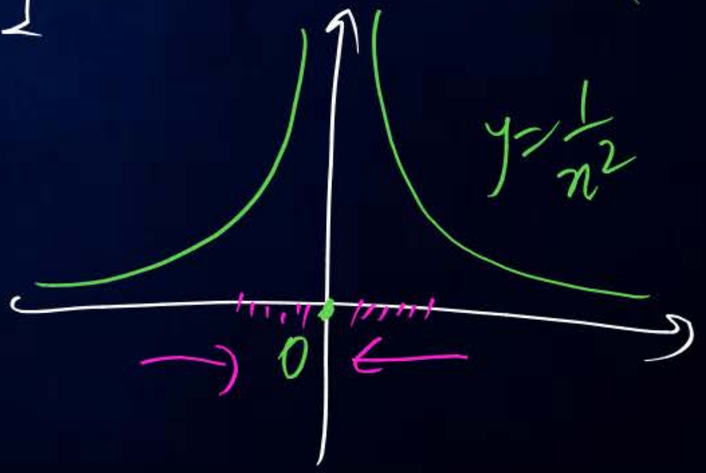
$= 1$ (unique)

(M-II) $\lim_{x\to 0}\left(\frac{\sin x}{x}\right) = \lim_{x\to 0}\left(\frac{\cos x}{1}\right) = \cos(0) = 1$

Qe $\lim_{x\to 0}\left(\frac{1}{x^2}\right) = ? = \frac{1}{0^2}$

$= +\infty$ = Not unique

= DNE




Thank
YOU